

Supplementary Information

Repositioning coal power to accelerate net-zero transition of China's power system

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Supplementary Note 1 Modeling framework

(1) CHEER framework

This study adopts a modeling framework focusing on the dynamic provincial-level power system transition pathways through soft-linking a multi-regional hybrid dynamic computable general equilibrium (CGE) model, named China Hybrid Energy Economic Research/CGE (CHEER/CGE)¹ and a multi-regional power system optimization model named CHEER/Power. The spatial resolution of both two models is provincial level and the timescale is from 2020 to 2060 with 5 years per period. CHEER/CGE model is introduced in its website (<https://www.cheer.org.cn/>). The version we adopt has 15 production sectors and 30 provinces of China, except for Tibet, Hongkong, Macau and Taiwan considering data availability. As a CGE model, CHEER/CGE represents the behavior of producers, consumers, government, investors and cross-regional traders based on general equilibrium theory, including profit and utility maximization, and zero profit. It also assumes the market equilibrium condition of all commodities and factors (i.e., labor and capital) and budgets balance of all agents. Under the policy shocks like carbon emissions cap, the price adjustment of commodities and factors will simultaneously reach a new general equilibrium status. The recursively dynamic mechanism of CHEER/CGE model represents the socioeconomic growth and energy demand as the results of four drivers, including capital accumulation, population change, natural resource supply and technical change. The model is adopted to project the provincial-level electricity demand from 2020 to 2060 under the socioeconomic development scenario and climate policies.

CHEER/Power model is an intertemporally dynamic provincial-level power system optimization model with 5 years per period, modeled as a mixed-integer linear programming (MILP) problem, coded in the Julia language, and solved using the Gurobi solver. CHEER/Power aims at optimizing the capacities and hourly dispatch of both generators, storage technologies, carbon capture and storage (CCS) and transmission lines, to minimizing the total discounted system costs till the target year with the satisfaction of a set of constraints, including the capacity constraints, electricity supply-demand balance in each time step of each period in each load zone, and operation condition of all generators. CHEER/Power model incorporates six modules as constraints, see Supplementary Note 2, i.e., power generation technology installation, retirement and retrofit decision module, unit commitment and economic dispatch module, energy storage investment and charge/discharge decision module, power transmission and distribution module, power system reserve module, and policy module. The basic model structure and formulation are referred to the Python-based open-source power system optimization model named SWITCH model²⁻⁵, but has three main updates. First, the JuMP programming framework in Julia language is adopted for CHEER/Power. The Julia-based modeling framework is more concise and faster. Second, CCS-retrofitting decisions of thermal power and early retirement decisions of coal power are endogenously represented in CHEER/Power. Third, the historical capacity data, more recent technology cost data and comprehensive resource potential estimation data are integrated into the model to reflect the reality of power system transition, see Supplementary Note 3.

(2) Projection of provincial-level electricity demand

We focus on projecting the provincial-level electricity demand trends from 2017 to 2060 under carbon neutrality goal based on CHEER model through the following steps. First, we set the gross domestic production (GDP) and population trends based on the shared socioeconomic pathways of China, which drives the production and electricity demand to a great extent. Second, we set the

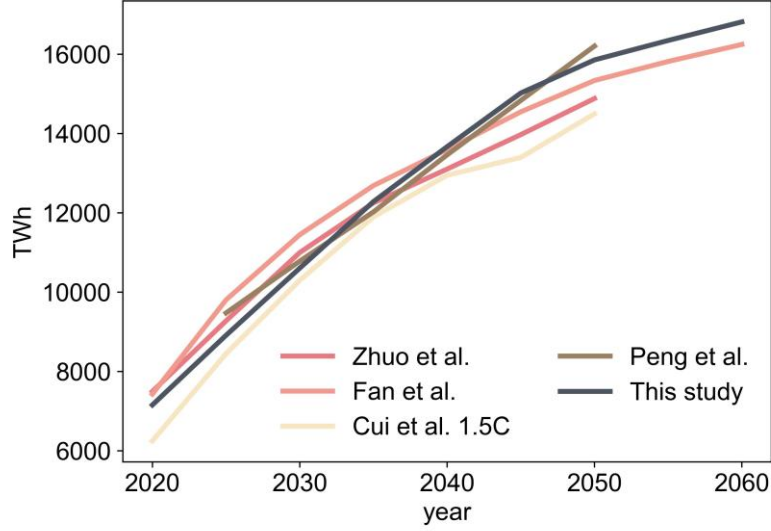
carbon neutrality pathways for non-electricity sector to CGE model, since the emissions of power sector are constraint by CHEER/Power model, and we only focus on the carbon neutrality pathways of non-electricity sector in CHEER model. Under the emissions cap, we consider the core electricity demand pathway to achieve the China's carbon neutrality goal. We assumed the growth rate of electricity demand in Tibet as the national average level, and the growth rates of Western and Eastern Inner Mongolia are assumed to be the same as the growth rate of Inner Mongolia in CHEER/CGE model. Based on the assumption above, we simulate the provincial-level electricity demand trend, shown in Supplementary Table 1. We compare the national electricity demand with the existing literatures shown in Supplementary Fig. 1, which shows that the demand scenario is consistent with the medium trends of existing literatures.

Supplementary Table 1 Provincial-level electricity demand scenario

Region	Province	2020	2025	2030	2035	2040	2045	2050	2055	2060
East	Anhui	229	263	301	351	401	452	483	499	516
North	Beijing	109	141	173	204	238	269	290	300	308
Southwest	Chongqing	114	133	149	164	171	177	178	177	176
East	Fujian	239	324	429	531	605	677	718	738	757
Northwest	Gansu	129	165	195	233	283	337	375	396	419
South	Guangdong	669	890	1140	1386	1582	1775	1913	2006	2088
South	Guangxi	195	219	241	262	279	295	302	305	306
South	Guizhou	146	161	179	195	207	217	220	222	223
South	Hainan	35	43	51	59	66	74	78	81	84
North	Hebei	372	469	554	643	706	764	794	807	819
North	Hebei	94	105	114	121	125	128	126	123	120
Northeast	Heilongjiang	316	379	445	521	572	621	645	655	665
Central	Hubei	201	232	254	278	301	322	330	329	328
Central	Hunan	179	217	256	305	349	393	418	433	446
East	Jiangsu	619	733	825	902	957	1001	1013	1014	1013
Central	Jiangxi	157	191	229	264	283	298	304	306	307
Northeast	Jilin	75	85	94	99	101	102	99	94	90
Northeast	Liaoning	232	267	300	331	360	384	391	393	395
Northwest	Ningxia	98	121	147	181	203	224	238	246	255
Northwest	Qinghai	71	90	110	130	148	165	176	182	186
Northwest	Shaanxi	162	195	233	274	323	373	403	423	441
North	Shandong	675	929	1138	1371	1467	1559	1602	1622	1650
East	Shanghai	152	201	242	285	332	378	408	432	454
North	Shanxi	220	263	310	362	409	457	492	511	530
Southwest	Sichuan	261	314	345	361	370	377	374	370	365
North	Tianjin	84	104	118	131	143	153	158	160	162
Southwest	Tibet	7	9	11	12	14	15	16	16	16
Northwest	Xinjiang	296	353	425	503	621	755	862	932	1004
South	Yunnan	183	218	250	282	287	290	288	284	278
East	Zhejiang	467	660	836	1004	1198	1405	1562	1690	1815
North	Western Inner	278	329	383	406	426	443	450	451	449

North	Mongolia										
	Eastern	Inner	95	113	131	139	146	151	154	154	153
	Mongolia										

Note: region refers to the power grid region in which the province is located. For simplicity, both the Eastern and Western Inner Mongolia are merged into the North China grid. Unit: TWh.



Supplementary Fig. 1 Projected electricity demand in three scenarios from 2020 to 2060. We compared the projected demand with four existing studies^{4,6-8}. TWh terawatt-hour.

Supplementary Note 2 Modules of CHEER/Power

(1) Technology installation, retirement, and retrofit decision module

Consider the newly installed capacity $INSC_{g,pb}$ for each power generation project g during the construction period pb . The installed capacities of various projects in historical periods prior to the initial simulation year p_0 , should be equal to the newly built capacities $INSC0_{g,pd}$ in each period, satisfying the requirements of supplementary equation (1). The installed capacities of all projects must be non-negative.

$$INSC_{g,pd} = INSC0_{g,pd} \text{ if } pd \leq p_0 \quad (1)$$

For power generation projects gr that allow early retirement, the newly retired capacity $newRT_{gr,pb,pt}$ and the cumulative early retired capacity $RT_{gr,pb,pt}$ in the retirement period pt during the operational life of projects constructed in period pb must both be non-negative and satisfy supplementary equation (2).

$$0 \leq RT_{gr,pb,pt} = RT_{gr,pb,ptlag} + newRT_{gr,pb,pt} \quad (2)$$

Here, $ptlag$ refers to the period before pt .

For power generation projects gr that allow CCS retrofitting (i.e., coal-fired, gas-fired, and biomass-fired power plants), the newly retrofitted CCS capacity $newCC_{gr,pb,pt}$ and the cumulative CCS retrofitted capacity $CC_{gr,pb,pt}$ in the retrofit period pt during the operational life of projects constructed in period pb must be non-negative and satisfy supplementary equation (3).

$$0 \leq CC_{gr,pb,pt} = CC_{gr,pb,ptlag} + newCC_{gr,pb,pt} \quad (3)$$

If retirement or CCS retrofitting occurred in historical periods, the cumulative scale must equal the exogenous scale in those periods.

In pt, the installed capacity of power generation project gr constructed in pb must exceed the cumulative retired and CCS-retrofitted capacity in pt, satisfying supplementary equation (4).

$$CC_{gr,pb,pt} + RT_{gr,pb,pt} \leq INSC_{gr,pb} \text{ if } pt < pb + \tau_{gr} \quad (4)$$

During simulation period p, the total capacity of power generation project g $Cap_{g,p}$, can be categorized into CCS retrofitted capacity $Cap_{g,p}^{CC}$ and non-CCS retrofitted capacity $Cap_{g,p}^{noCC}$. The relationship with installed, retired, and retrofitted capacity are expressed in supplementary equations (5-7). For projects without CCS retrofitting, $CC_{gr,pb,p}$ is 0, and for projects without retirement, $RT_{gr,pb,p}$ is 0. The period must be within the life expectancy of given power generator, i.e., $p < pb + \tau_g$, here τ_g is the life expectancy.

$$Cap_{g,p}^{CC} = \sum_{p-\tau_g \leq pb \leq p} CC_{g,pb,p} \quad (5)$$

$$Cap_{g,p}^{noCC} = \sum_{p-\tau_g < pb \leq p} INSC_{g,pb} - \sum_{p-\tau_g < pb \leq p} RT_{g,pb,p} - \sum_{p-\tau_g < pb \leq p} CC_{g,pb,p} \quad (6)$$

$$Cap_{g,p} = Cap_{g,p}^{CC} + Cap_{g,p}^{noCC} \quad (7)$$

The projects of solar photovoltaics (PV), wind and hydropower are subject to upper limits, i.e., maximum installation potentials $maxCap_g$, thus satisfying the requirements of supplementary equation (8).

$$Cap_{g,p} \leq maxCap_g \quad (8)$$

During simulation period p, the capital costs $Capex_{g,p}$ and fixed O&M costs $FixOpex_{g,p}$ of power generation project g are the sum of the project capacities built during pb, multiplied by the unit annualized capital cost and fixed O&M cost in the respective period, as shown in supplementary equations (9-11). The fixed cost of CCS retrofitting is the product of retrofitted capacities and the unit annualized retrofitting cost during pt. Retired projects do not incur fixed O&M costs. The total fixed cost of power generation is expressed in supplementary equation (12). The annualized capital cost is related to the discount rate, as shown in supplementary equation (13).

$$Capex_{g,p} = \sum_{p-\tau_g < pb \leq p} INSC_{g,pb} \times CapCost_{g,pb} \quad (9)$$

$$FixOpex_{g,p} = \sum_{p-\tau_g < pb \leq p} (INSC_{g,pb} - RT_{g,pb,p}) \times fixOMCost_{g,pb} \quad (10)$$

$$CCCapex_{gr,p} = \sum_{p-\tau_g < pb \leq p} \sum_{pb \leq pt \leq p} CC_{gr,pb,pt} \times fixCCCost_{g,pb,pt} \quad (11)$$

$$FixGen_p = \sum_g Capex_{g,p} + FixOpex_{g,p} + CCCapex_{gr,p} \quad (12)$$

$$CapCost_{g,pb} = Overnight_{g,pb} \frac{ir}{(1 - (1 + ir)^{-\tau_g})} \quad (13)$$

where, $Overnight_{g,pb}$ represents the overnight investment cost of project g during period pb, ir is the discount rate.

(2) Unit commitment and economic dispatch module

At timepoint t, the dispatchable capacity of project g with CCS and without CCS retrofitting are $DP_{g,t}^{CC}$ and $DP_{g,t}^{noCC}$, respectively, and both are non-negative variables. Considering the energy penalty $loss_g^{CC}$ of CCS retrofitted projects, the total dispatchable capacity $DP_{g,p}$ satisfies supplementary equation (14).

$$DP_{g,t} = DP_{g,t}^{noCC} + DP_{g,t}^{CC} \times (1 - loss_g^{CC}) \quad (14)$$

The total dispatchable capacity in region r at timepoint t $DPReg_{r,t}$ is the sum of all project

capacities, as shown in supplementary equation (15).

$$\text{DPR}_{g,t} = \sum_{R(g) \in r} \text{DP}_{g,t} \quad (15)$$

For thermal power, the fuel consumption for CCS retrofitted, non-CCS retrofitted, and total dispatch at time t are $\text{Fuel}_{g,t,f}^{\text{CC}}$, $\text{Fuel}_{g,t,f}^{\text{noCC}}$, and $\text{Fuel}_{g,t,f}$, respectively. Here, f represents the fuel type corresponding to the thermal power technology. The carbon emissions at time t $\text{Emit}_{g,t,f}$ are given in supplementary equation (16):

$$\text{Emit}_{g,t,f} = \text{Fuel}_{g,t,f}^{\text{noCC}} \times \text{EF}_f + \text{Fuel}_{g,t,f}^{\text{CC}} \times \text{EF}_f \times (1 - \kappa_g) - \text{Fuel}_{g,t,f} \times \text{upEF}_f \quad (16)$$

where κ_g is the CCS capture leakage rate, EF_f is the emission factor of fuel f , and upEF_f is the upstream emission factor. The upstream emissions of biomass are negative.

The total annual emissions Temit_p are the sum of emissions across all timepoints, fuels and projects, as shown in supplementary equation (17):

$$\text{Temit}_p = \sum_{g,p(t)=p,f} \text{Emit}_{g,t,f} \times \text{num}_t \quad (17)$$

where, num_t is the number of each representative timepoint in one year.

Variable O&M costs of power generation VarOpex_t and CCS retrofitting projects CCVarOpex_t are given in supplementary equations (18—19):

$$\text{VarOpex}_t = \sum_g \text{DP}_{g,t} \times \text{varOM}_g \quad (18)$$

$$\text{CCVarOpex}_t = \sum_g \text{DP}_{g,t}^{\text{CC}} \times (\text{varOM}_g^{\text{CC}} + \text{EF}_f \times \kappa_g \times (\text{trans}_g^{\text{CC}} + \text{strg}_g^{\text{CC}})) \quad (19)$$

where, varOM_g and $\text{varOM}_{g,f}^{\text{CC}}$ are the annual variable O&M costs of generation projects and carbon capture facilities, respectively; $\text{trans}_g^{\text{CC}}$ and $\text{strg}_g^{\text{CC}}$ are the carbon transport and storage costs of each project g , respectively.

For unit commitment, the CCS and non-CCS committed capacities of project g at time t are $\text{Comm}_{g,t}^{\text{CC}}$ and $\text{Comm}_{g,t}^{\text{noCC}}$, respectively. The total committed capacity $\text{Comm}_{g,t}$ is the sum of above two variables, as shown in supplementary equation 20. The maximum committed capacity is the product of installed capacity and availability rate, while nuclear power projects are constrained by minimum commitment requirements to ensure base-load operation.

$$\text{Comm}_{g,t} = \text{Comm}_{g,t}^{\text{CC}} + \text{Comm}_{g,t}^{\text{noCC}} \quad (20)$$

Through start-up and shut-down of power units, the committed capacity changes. This study simplified the unit commitment issue as a linear problem. The start-up capacity and shut-down capacity of project g at time t are $\text{Start}_{g,t}$ and $\text{Shut}_{g,t}$, respectively. The relationship between the committed capacity at time t and tag is described in supplementary equation (21).

$$\text{Comm}_{g,t} = \text{Comm}_{g,\text{tag}} + \text{Start}_{g,t} - \text{Shut}_{g,t} \quad (21)$$

The start-up cost StartOpex_t at time t is the sum of the products of the start-up capacity over 1 hour for each unit and the unit start-up operational cost startOM_g , as shown in supplementary equation (22).

$$\text{StartOpex}_t = \sum_g (\text{Start}_{g,t} \times \text{startOM}_g / \text{dur}) \quad (22)$$

Where dur is the hours represented by each timepoint.

Thermal power projects have minimum start-up and shut-down time. At time t , the committed capacity must exceed the sum of all capacities that start within the minimum start-up time, and must be less than the sum of the maximum committed capacity and all capacities that shut down within

the minimum shut-down time, as shown in supplementary equations (23—24).

$$\text{Comm}_{g,t} \geq \sum_{t-\text{start} \leq i \leq t} \text{Start}_{g,i} \quad (23)$$

$$\text{Comm}_{g,t} \leq \text{avail}_g \times \text{Cap}_{g,p(t)} - \sum_{t-\text{shut} \leq i \leq t} \text{Shut}_{g,i} \quad (24)$$

The relationship between dispatchable capacity and committed capacity is determined by the load fraction. For thermal and nuclear power, there is a minimum load fraction $\text{ldf}_g^{\min}$, while for renewable energy, the minimum load fraction is 0. Additionally, renewable energy generation has a maximum output load factor $\text{ldf}_g^{\max}$, as shown in supplementary equations (25).

$$\text{ldf}_g^{\min} \times \text{Comm}_{g,t}^{\text{CC/noCC}} \leq \text{DP}_{g,t}^{\text{CC/noCC}} \leq \text{ldf}_g^{\max} \times \text{Comm}_{g,t}^{\text{CC/noCC}} \quad (25)$$

Fuel use is directly related to dispatchable capacity, see supplementary equation (26). The fuel cost $\text{FuelCost}_{r,p,f}$ is related to fuel use and fuel price, see supplementary equation (27). Here, the maximum biomass use of each region is constraint by the biomass potential.

$$\text{Fuel}_{g,t,f}^{\text{CC/noCC}} \leq \text{fuelr}_g \times \text{DP}_{g,t}^{\text{CC/noCC}} \quad (26)$$

$$\text{FuelCost}_p = \sum_f \sum_{p(t)=p} \sum_{R(g)=r} \text{Fuel}_{g,t,f} \times \text{fuelprice}_{r,p,f} \quad (27)$$

(3) Energy storage investment and charge/discharge decision module

Consider the newly added energy capacity $\text{INSE}_{g,pb}$ for each energy storage project g during construction period pb . The energy capacities of various projects in historical periods before the initial simulation year p_0 are considered stock. Therefore, they should equal the newly built capacity $\text{INSE0}_{g,pd}$ in each period, satisfying supplementary equation (28).

$$\text{INSE}_{g,pd} = \text{INSE0}_{g,pd} \text{ if } pd \leq p_0 \quad (28)$$

In period p , the capital cost of the energy storage project g , denoted as $\text{EneCapex}_{g,p}$, is the sum of the energy capacity built during each construction period pb multiplied by the unit annualized capital cost $\text{EneCapCost}_{g,pb}$ in the corresponding period, as shown in supplementary equation (29). $\text{EneCapCost}_{g,pb}$ is the annualized value of the overnight capital cost per unit of energy capacity, like supplementary equation (13).

$$\text{CapexSTR}_p = \sum_g \sum_{p-\tau_g < pb \leq p} \text{INSE}_{g,pb} \times \text{EneCapCost}_{g,pb} \quad (29)$$

The energy capacity of energy storage project g in period p , denoted as $\text{EneCap}_{g,p}$ is given by supplementary equation (30).

$$\text{EneCap}_{g,p} = \sum_{p-\tau_g < pb \leq p} \text{INSE}_{g,pb} \quad (30)$$

At time t , the charging power is $\text{CG}_{g,t}$. The total charging power for a region is $\text{CGReg}_{r,t}$, as shown in supplementary equation (31).

$$\text{CGReg}_{r,t} = \sum_{R(g) \in r} \text{CG}_{g,t} \quad (31)$$

If the discharge duration CGDur_g is exogenously determined, the energy storage power $\text{DP}_{g,p}$ and energy capacity $\text{CG}_{g,p}$ satisfy the relationship given in supplementary equation (32).

$$\text{CG}_{g,t} = \text{DP}_{g,t} \times \text{CGDur}_g \quad (32)$$

The charging power should not exceed the maximum available power, meaning the condition in supplementary equation (33) must be satisfied.

$$CG_{g,t} \leq \text{avail}_g \times \text{Cap}_{g,p(t)} \quad (33)$$

At time t , the stored energy is denoted by $\text{STR}_{g,t}$. The relationship between stored energy, charging power, and discharge power is given by supplementary equation (34). In which, the discharge power is equal to the power multiplied by dur .

$$\text{STR}_{g,t} = \text{STR}_{g,\text{tlag}} + CG_{g,t} \times \text{eff}^{\text{str}} - DP_{g,t} \times \text{dur} \quad (34)$$

Stored energy must be less than the energy capacity, which is expressed by the condition in supplementary equation (35).

$$\text{STR}_{g,t} \leq \text{EneCap}_{g,p(t)} \quad (35)$$

If there is a maximum number of cycles num^{iter} per period, then the total discharge power must not exceed the energy capacity under the maximum cycle count, as shown in supplementary equation (36):

$$\sum_{p(t)=p} DP_{g,t} \times \text{dur} \leq \text{num}^{\text{iter}} \times \text{EneCap}_{g,p(t)} \quad (36)$$

(4) Power transmission and distribution module

Consider the construction capacity $\text{INST}_{l,\text{pb}}$ of each transmission line l during the construction period pb . The total capacity in period p , $\text{TmCap}_{g,p}$, is given by supplementary equation (37):

$$\text{TmCap}_{g,p} = \sum_{\text{pb} \leq p} \text{INST}_{l,\text{pb}} + \text{INST0}_l \quad (37)$$

where INST0_l is the transmission capacity already built in the initial year of the study period.

The fixed transmission costs FixedTM_p include capital costs TmCapCost_l and fixed O&M costs TmfixOMCost_l , as shown in supplementary equation (38):

$$\text{FixedTM}_p = \sum_l \text{TmCap}_{l,p} \times (\text{TmCapCost}_l + \text{TmfixOMCost}_l) \quad (38)$$

Each transmission line l can perform bi-directional transmission, meaning that each line corresponds to two groups of output and input provinces, denoted as (re, rm) , where both re and rm are interchangeable. Thus, the number of province clusters is twice the number of transmission lines. If the transmission power at time t is $\text{DPT}_{\text{re},\text{rm},t}$, the transmission power limit is the transmission capacity of the line used, as shown in supplementary equation (39):

$$\text{DPT}_{\text{re},\text{rm},t} \leq \text{TmCap}_{l(\text{re},\text{rm}),p(t)} \quad (39)$$

At time t , the net import amount for a province r , denoted as $\text{NPT}_{\text{rm},t}$, is given by supplementary equation (40), where eff^{tm} is the transmission efficiency:

$$\text{NPT}_{r,t} = \sum_{\text{re}} \text{DPT}_{\text{re},r,t} \times \text{eff}^{\text{tm}} - \sum_{\text{rm}} \text{DPT}_{r,\text{rm},t} \quad (40)$$

Consider the construction capacity $\text{INSD}_{r,\text{pb}}$ for each region's distribution lines during the construction period pb . The total capacity in period p , $\text{DsCap}_{r,p}$, is given by supplementary equation (41):

$$\text{DsCap}_{r,p} = \sum_{\text{pb} \leq p} \text{INSD}_{r,\text{pb}} + \text{INSD0}_r \quad (41)$$

where INST0_l is the regional transmission and distribution capacity already completed in the initial year of the study period.

The fixed costs for regional distribution lines are expressed in supplementary equation (42):

$$\text{FixedDS}_p = \sum_r \text{DsCap}_{r,p} \times \text{DsfixCost}_l \quad (42)$$

The distribution capacity for each region must exceed the expected peak load for that region to

ensure grid safety, as shown in supplementary equation (43), where eff^{ds} is the distribution efficiency (i.e., accounting for line losses):

$$\text{DsCap}_{r,p} \times \text{eff}^{\text{ds}} \geq \text{Peak}_{r,p} \quad (43)$$

For each region r , the total dispatch capacity $\text{DPReg}_{r,t}$ and net input power $\text{NPT}_{r,t}$ at time t together provide the total supply for the regional grid, which is used for storage charging $\text{CGReg}_{r,t}$ and to meet demand $\text{load}_{r,t}$. This relationship is given by supplementary equation (44):

$$\text{DPReg}_{r,t} + \text{NPT}_{r,t} = \text{CGReg}_{r,t} + \text{load}_{r,t} / \text{eff}^{\text{ds}} \quad (44)$$

(5) Power system reserve module

The spinning reserve capacity of each power generation project g at timepoint t is denoted as $\text{Spin}_{g,t}$. For energy storage facilities, the reserve can be provided by either increasing discharge or reducing charging power, as shown in supplementary equation (45). Additionally, there is a requirement for 10-minute spinning reserve, so the spinning reserve capacity is constrained by the stored energy, as shown in supplementary equation (46):

$$\text{Spin}_{g,t} \leq \text{avail}_g \times \text{Cap}_{g,p(t)} - \text{DP}_{g,t} + \text{CG}_{g,t} \quad (45)$$

$$\text{Spin}_{g,t} \leq 6 \times \text{STR}_{g,t} \quad (46)$$

For non-storage projects, spinning reserve can be provided by increasing output, as shown in supplementary equation (47), with the maximum reserve limited by the ramp-up rate ramp_g , as shown in supplementary equation (48):

$$\text{Spin}_{g,t} \leq \text{avail}_g \times \text{Cap}_{g,p(t)} - \text{DP}_{g,t} \quad (47)$$

$$\text{Spin}_{g,t} \leq \text{ramp}_g \times \text{avail}_g \times \text{Cap}_{g,p(t)} \quad (48)$$

The total spinning reserve for each region is denoted as $\text{SpinReg}_{r,t}$, as shown in supplementary equation (49):

$$\text{SpinReg}_{r,t} = \sum_{R(g)=r} \text{Spin}_{g,t} \quad (49)$$

Spinning reserve is used to meet unexpected incidents, fluctuations in renewable energy, and forecast errors in demand. Therefore, the following constraint must be satisfied, as shown in supplementary equation (50):

$$\sum_{R(g)=r} \alpha_g \times \text{DP}_{g,t} + \beta \times \frac{\text{load}_{r,t}}{\text{eff}^{\text{ds}}} \leq \text{SpinReg}_{r,t} \quad (50)$$

where, α_g is the failure rate or prediction error rate for power generation projects, which is set as 30% for VRE and 10% for others. β is the load forecasting error, set to 5%.

Planned reserve is provided by generation and energy storage devices. At each time point, the planned reserve supply from energy storage devices (i.e., STR) is the net discharge, and the planned reserve supply from variable renewable energy is determined by its output curve. The planned reserve for non-variable renewable energy is simply its capacity. The planned reserve ratio (pr_r) needs to reach 15%, meaning it must be able to meet an additional 15% of demand load, as shown in supplementary equation (51):

$$\sum_{R(g)=r \text{ \& } g \in \text{STR}} (\text{DP}_{g,t} - \text{CG}_{g,t}) + \sum_{R(g)=r \text{ \& } g \notin \text{STR}} \text{ldf}_g^{\text{max}} \times \text{Cap}_{g,p(t)} \geq (1 + \text{pr}_r) \times \frac{\text{load}_{r,t}}{\text{eff}^{\text{ds}}} \quad (51)$$

(6) Policy module

The model in this study firstly considers the carbon cap policy, CO2Cap_p , restricts the total emissions, as shown in supplementary equation (52):

$$\text{Temit}_p \leq \text{CO2Cap}_p \quad (52)$$

The model also considers planning policies for power generation and transmission projects. The total capacity could have an upper or lower constraints according to the research needs. Here, the nuclear power capacity and transmission line capacity are constraint by the upper limit, while the pumped hydro is constraint by the lower limit, to reflect the national energy planning target.

(7) Optimization objective

The optimization objective of the model is to minimize the discounted total system cost (TotSysCost), where the total cost per period (SysCost_p) is shown in supplementary equation (53), and the total discounted system cost is shown in supplementary equation (54):

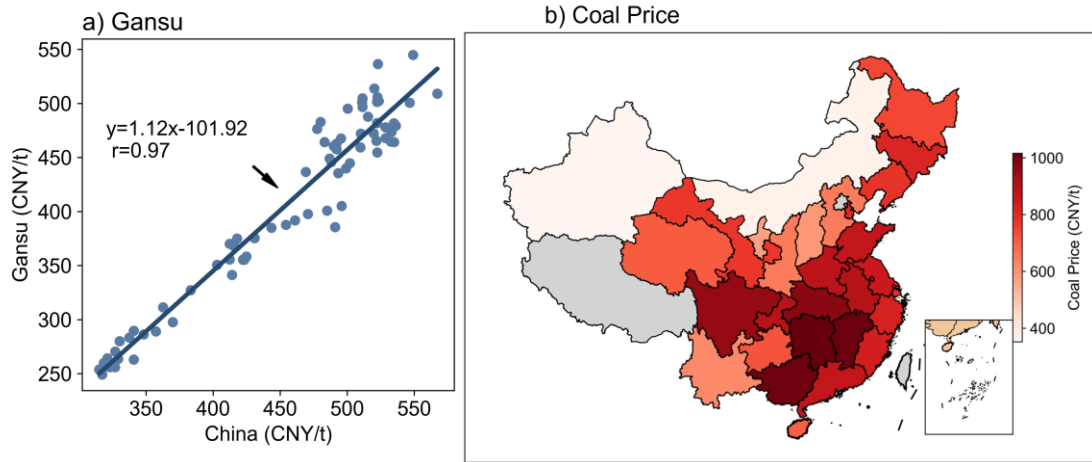
$$\text{SysCost}_p = \text{CapexGen}_p + \text{CapexSTR}_p + \text{FuelCost}_p + \text{FixedTM}_p + \text{FixedDS}_p + \text{EmisCost}_p + \sum_{p(t)=p} (\text{VarOpex}_t + \text{CCVarOpex}_t + \text{StartOpex}_t) \times \text{num}_t \quad (53)$$

$$\text{TotSysCost} = \sum_p \text{SysCost}_p \times (1 + \text{ir})^{-(p-2020)} \quad (54)$$

Supplementary Note 3 Data preparation for CHEER/Power

(1) Fuel costs

The National Development and Reform Commission (NDRC) Price Monitoring Center of China has been releasing national and provincial coal price indices since 2014 to reflect the ex-factory prices of coal nationwide and in each province. The monitoring covers 30 provinces and municipalities excluding Tibet and can be found at <http://meeting.baiinfo.com/dianmei>, providing a relatively accurate reflection of fuel costs for coal-fired power plants. However, this index ceased publication after January 2020. Subsequently, the China Electricity Council (CEC) began publishing unified national coal indices since May 2019, available at <https://www.cec.org.cn/dmzs/index.html>. Among these, the Caofeidian index, based on cost data at the Caofeidian port, provides the most comprehensive reflection of coal prices for power generation. In recent years, coal prices in China have continued to rise, with an average annual increase of approximately 82 Chinese Yuan (CNY) per ton for 5000 kilocalorie (kcal) coal, rising from an average price of 513 CNY per ton in 2020 to 786 CNY per ton in 2023. Considering the continuous fluctuations in coal prices, it is assumed that future coal prices will remain at the 2023 level. To calculate the coal prices in each province, a linear regression analysis was conducted on the provincial price indices and the national price index from 2014 to 2020. This was used to infer the coal prices for each province in 2023. The fitting effect is demonstrated using Gansu provinces as examples (Supplementary Fig. 2a). Regression results indicate that, except for Inner Mongolia and Xinjiang, where coal resources are abundant and thus less affected by coal supply, the determination coefficients for the remaining provinces are all above 0.8, with the majority exceeding 0.95. This demonstrates the credibility of this method. To align with other fuels, prices are converted to unit heat value prices in U.S. Dollars (USD, \$) per gigajoule (GJ), with coal prices and regression coefficients shown in the first two columns of Supplementary Table 2 and Supplementary Fig. 2b.



Supplementary Fig. 2 Estimated provincial-level coal price based on linear regression. a the regression case of Gansu between provincial-level coal price and national price; **b.** the estimated coal price in 2023. r is the correlation coefficient between the coal price of Gansu province (y) and national average price (x).

For natural gas, due to the lack of real-time price indices for each province in China, the benchmark gate station prices for natural gas in each province published by the NDRC in 2018 are used as a reference (see

<https://www.gov.cn/xinwen/2019-03/29/5378081/files/c5c16bd8d91a46c8a7a3b65fbc597f2a.pdf>).

An increase of 0.8 CNY/m³ above the actual situation is assumed for the ex-factory price, and the units are converted and placed in the third column of Supplementary Table 2. For biomass fuels, the ex-factory price is determined based on the evaluation by Yuan et al.⁹. As biomass fuel should not significantly occupy arable land and forest land, the limited potential of biomass resources is acknowledged. The maximum potential of different agricultural and forestry residues and energy crops in each province was evaluated based on the assessment by Wang et al.¹⁰. Biomass fuel prices and resource potential limits are shown in the fourth and fifth columns of Supplementary Table 2.

Supplementary Table 2 Provincial-level fuel costs and biomass potential

Region	Coal	R Value	Gas	Biomass	Biomass Cap
Unit	\$/GJ		\$/GJ	\$/GJ	EJ
Anhui	6.11	0.99	10.25	4.69	1.66
Beijing			9.92	5.41	0.13
Chongqing	6.09	0.97	8.62	4.68	0.71
East Inner Mongolia	2.63	0.84	7.51	4.75	0.87
Fujian	5.71	0.97	10.58	5.01	1.19
Gansu	5.29	0.97	7.84	4.51	0.97
Guangdong	5.93	0.98	10.58	5.11	1.26
Guangxi	6.85	0.96	9.95	4.65	2.61
Guizhou	4.95	0.94	8.88	4.49	1.97
Hainan	4.80	0.88	8.62	4.94	0.24
Hebei	4.40	0.94	9.84	4.86	1.73
Heilongjiang	5.09	0.83	9.06	4.81	1.55

Henan	6.01	0.98	9.95	4.72	2.11
Hubei	6.60	0.99	9.77	4.84	2.26
Hunan	6.93	0.97	9.77	4.76	2.20
Jiangsu	5.81	0.99	10.51	5.23	1.23
Jiangxi	6.89	0.99	9.77	4.82	1.77
Jilin	5.58	0.88	9.06	4.84	1.34
Liaoning	5.36	0.95	9.84	4.94	1.13
Ningxia	3.92	0.95	7.25	4.57	0.26
Qinghai	4.85	0.90	8.14	4.66	0.09
Shaanxi	4.43	0.90	7.51	4.57	1.54
Shandong	5.93	0.97	9.84	4.96	2.12
Shanghai	5.44	0.97	10.58	5.28	0.08
Shanxi	4.03	0.99	9.58	4.82	1.33
Sichuan	6.49	0.98	8.66	4.70	2.53
Tianjin	5.51	0.98	9.92	5.41	0.13
Tibet			10.58	4.95	0.58
West Inner Mongolia	2.49	0.76	7.51	4.75	0.87
Xinjiang	2.40	0.70	6.81	4.95	0.73
Yunnan	4.23	0.75	8.88	4.54	3.51
Zhejiang	5.63	0.94	10.54	5.28	0.75

Note: Coal, Gas, and Biomass indicate the fuel price in each province. Biomass Cap is the bioenergy potential in each province. R Value is the correlation coefficient between the coal price of each province and the national average price. \$/GJ dollars per gigajoule. EJ exajoule.

(2) Solar and wind potential

Wind and solar energy are important types of energy resources that can be converted into electricity through wind turbines and photovoltaic systems. Our model distinguishes between four types of wind and solar resources: onshore wind, offshore wind, central PV, and distributed PV. The wind and solar resource potential vary significantly among provinces and regions due to two main factors: the maximum installed capacity and the differences in utilization hours resulting from variations in wind speed and solar irradiance intensity.

Numerous studies have assessed the wind and solar resource potential in various regions of China and have widely acknowledged that these resources significantly exceed existing electricity demand^{11–13}. It is necessary to carefully consider factors such as land use type, natural reserves, distance from urban areas, and slope when selecting suitable areas. We firstly evaluate the available land area for onshore wind and central PV. The China Land Use/Cover Change Dataset for 2020 provided by the Resource and Environmental Science and Data Center of the Chinese Academy of Sciences (RESDC)¹⁴ is adopted to represent the grid-level land use and cover. Natural conditions need to be met first, including slope, altitude, and nature reserves, as shown in Supplementary Table 3. The utilization factors for onshore wind power vary with different slopes; we referred to Wang et al.'s study to set the utilization ratios for different slopes¹¹. The utilization rates for different land use types also vary; we referenced previous studies to determine the available area ratios for onshore wind power and central PV in low, medium, and high potential assessments, as shown in Supplementary Table 4^{11,15,16}. Distributed PV is primarily distributed in urban areas, and we assumed that the available urban area for distributed PV would be 10%, 15%, and 20% in three

different scenarios. For offshore wind power, this study focuses on the nearshore wind, which uses fixed-bottom wind turbines and locates 10-70 kilometers (km) from the coastline in waters no deeper than 50 meters. The grid map for the spatial distribution of wind and solar PV is shown in Supplementary Fig. 3.

Supplementary Table 3 Natural resource conditions for onshore wind and central PV

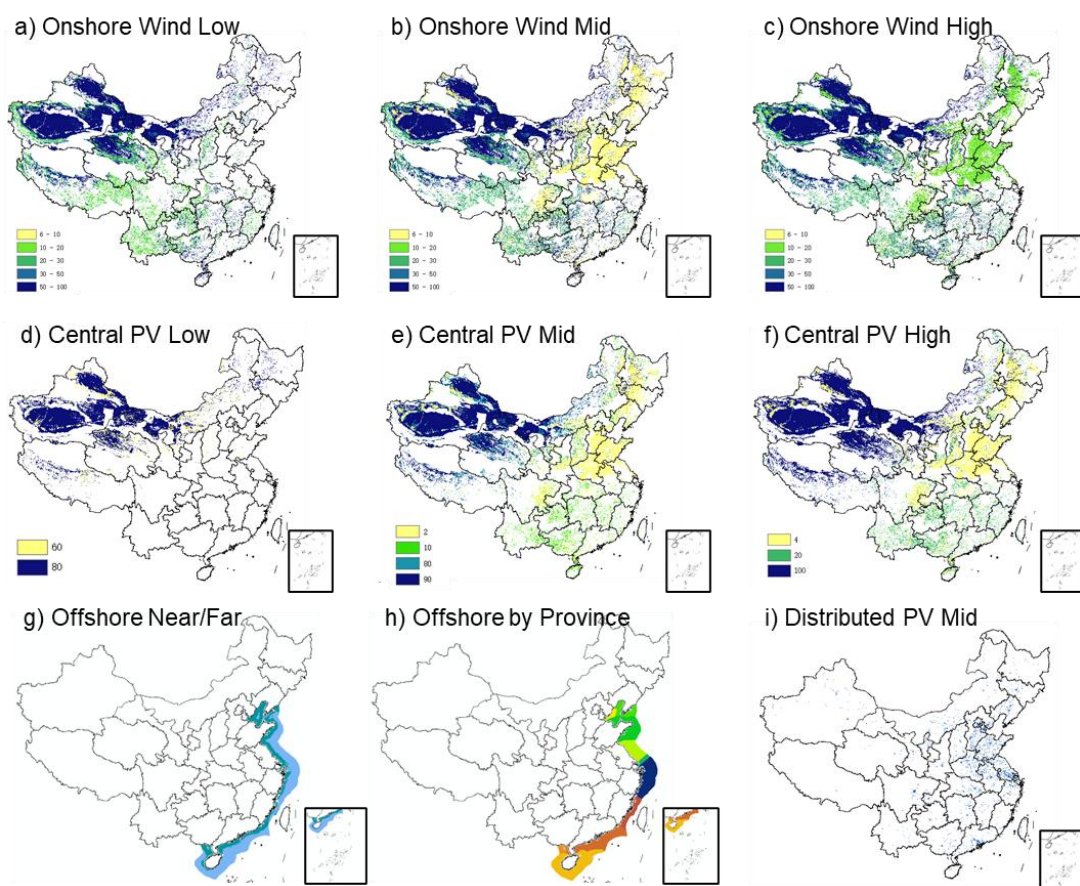
	Onshore Wind			Central PV		
	Altitude	Natural reserve	Slope	Altitude	Natural reserve	Slope
Low	≤5000	0	<3%: UF=1 3-6%: UF=0.5	≤5000	0	≤3%
Mid	≤5000	0	6-30%: UF=0.3	≤5000	0	≤5%
High	≤5000	0	≥30%: UF=0	≤5000	0	≤7%

Note: Low, Mid, and High indicate the low, medium, and high potential estimation scenarios. UF the utilization factor of onshore wind, PV solar photovoltaic technology.

Supplementary Table 4 Land availability for onshore wind and central PV

Land type	Onshore Wind			Central PV		
	Low	Mid	High	Low	Mid	High
Paddy cropland	0	0	0	0	0	0
Dry cropland	0	10	20	0	2	5
Closed forest land	0	0	0	0	0	0
Shrubland	60	80	100	10	20	30
Open forest land	60	80	100	10	20	30
Other forest land	60	80	100	10	20	30
High coverage grassland	0	0	0	0	0	0
Moderate coverage grassland	0	0	0	0	0	0
Low coverage grassland	60	80	100	60	80	100
Urban and built-up land	0	0	0	0	0	0
Unused land	80	90	100	80	90	100

Note: Low, Mid, and High indicate the low, medium, and high potential estimation scenarios. PV indicates solar photovoltaic technology.



Supplementary Fig. 3 Land availability for onshore and offshore wind, central and distributed PV. Low, Mid, and High indicate the low, medium, and high potential estimation scenarios. **a-f** represent the land availability for onshore wind and central PV in low, medium, and high potential scenarios, and the legend is the proportion of available land, unit: %. **g** shows the offshore distance. Green grids are nearshore wind, and blue grids are far offshore wind. **h** shows the nearest provinces of each offshore grid. **i** shows the spatial availability of distributed PV. PV solar photovoltaic technology.

Based on the available area and the installation capacity per unit area, we can estimate the maximum available installation potential for each province under low, medium, and high scenarios. The unit installation capacities for onshore wind power, offshore wind power, central PV, and distributed PV are set at 5, 6, 90, and 90 megawatts (MW) per km², respectively^{13,15}. The estimated installation potential for each technology in each scenario by province is shown in Supplementary Table 5 and Supplementary Table 6.

Supplementary Table 5 Available maximum wind power capacity potential by province

Gigawatt (GW)	Onshore Wind			Offshore Wind
Province	Low	Mid	High	Near
Anhui	15.2	40.9	67.6	0.0
Beijing	3.8	6.5	9.8	0.0
Chongqing	23.1	31.2	44.9	0.0
East Inner Mongolia	222.4	282.3	361.7	0.0

Fujian	63.1	82.9	105.7	102.3
Gansu	512.3	589.6	685.2	0.0
Guangdong	75.5	97.4	129.1	265.4
Guangxi	168.8	214.5	278.5	67.8
Guizhou	114.6	138.0	177.4	0.0
Hainan	29.1	35.8	47.4	83.3
Hebei	26.3	69.1	117.5	81.5
Heilongjiang	137.0	193.3	262.4	0.0
Henan	14.6	65.7	120.6	0.0
Hubei	97.2	135.8	181.6	0.0
Hunan	134.2	173.5	222.4	0.0
Jiangsu	5.2	18.8	33.6	168.4
Jiangxi	99.0	128.4	164.7	0.0
Jilin	63.3	89.5	135.1	0.0
Liaoning	47.7	71.8	113.9	188.9
Ningxia	50.4	57.9	74.0	0.0
Qinghai	798.4	951.6	1111.2	0.0
Shaanxi	42.4	61.1	88.9	0.0
Shandong	22.4	79.4	146.0	333.1
Shanghai	0.3	0.5	0.6	31.3
Shanxi	81.4	103.9	149.4	0.0
Sichuan	163.4	217.2	297.1	0.0
Tianjin	1.5	4.6	8.0	5.5
Tibet	606.3	732.7	861.4	0.0
West Inner Mongolia	1068.0	1250.6	1460.7	0.0
Xinjiang	3169.2	3649.8	4153.8	0.0
Yunnan	213.7	266.1	342.4	0.0
Zhejiang	21.8	29.2	37.5	145.4

Note: Low, Mid, and High indicate the low, medium, and high potential estimation scenarios.

Supplementary Table 6 Available maximum solar power capacity potential by province

Gigawatt (GW)	Central PV			Distributed PV		
Province	Low	Mid	High	Low	Mid	High
Anhui	32.9	169.2	327.5	170.0	257.1	344.2
Beijing	5.4	29.0	64.6	39.0	60.9	83.1
Chongqing	36.5	141.1	299.5	18.4	32.5	47.2
East Inner Mongolia	2555.3	3549.5	4565.4	36.0	57.7	78.1
Fujian	112.8	418.4	879.0	50.1	94.1	140.6
Gansu	8115.5	10393.4	12726.7	24.4	46.3	68.5
Guangdong	202.6	466.4	793.4	129.1	204.5	279.3
Guangxi	429.0	1123.0	1924.0	70.7	114.4	156.8
Guizhou	220.5	787.1	1547.0	16.3	32.1	46.6
Hainan	96.2	168.3	258.4	16.9	25.7	34.4
Hebei	119.7	353.5	655.0	194.0	297.5	400.1

Heilongjiang	1610.1	1965.8	2477.0	30.9	47.7	64.2
Henan	35.0	270.5	543.9	234.0	359.0	482.2
Hubei	241.0	642.8	1171.1	98.5	150.7	202.9
Hunan	333.4	850.9	1469.8	62.8	101.4	138.9
Jiangsu	23.4	79.7	141.5	252.5	380.5	507.8
Jiangxi	247.0	615.1	1051.0	52.2	84.2	115.2
Jilin	972.1	950.8	1206.6	65.9	101.2	135.6
Liaoning	245.9	402.3	642.1	120.9	190.8	256.8
Ningxia	785.9	1002.4	1279.9	24.1	37.2	50.2
Qinghai	10896.4	16747.5	23277.2	14.8	27.1	39.4
Shaanxi	242.0	433.6	708.7	37.0	61.7	84.6
Shandong	175.7	435.8	752.2	309.8	478.6	642.5
Shanghai	1.1	2.3	3.4	23.0	34.5	46.0
Shanxi	564.4	1095.4	1781.4	69.4	120.2	167.9
Sichuan	261.9	1059.6	2368.3	63.6	101.2	139.6
Tianjin	25.7	37.3	51.7	35.3	52.9	70.5
Tibet	6399.1	11345.0	16653.1	3.3	6.3	9.4
West Inner Mongolia	18814.3	22284.3	25897.9	111.9	175.5	237.0
Xinjiang	53073.7	64886.4	75983.7	61.9	96.0	128.8
Yunnan	253.9	1074.5	2439.6	26.5	52.3	79.6
Zhejiang	47.0	134.6	249.7	95.1	154.0	212.1

Note: Low, Mid, and High indicate the low, medium, and high potential estimation scenarios. PV indicates solar photovoltaic technology.

To evaluate the output curves for different technologies in each province, we utilized the global wind and solar output database provided by <https://renewables.ninja/>. We randomly selected 100 representative points for onshore wind power and 50 representative points for other technologies to derive the output curves for each province. Given the significant deviation of the PV capacity factor from actual data, we used the Global Solar Atlas database to calculate the provincial average utilization hours. The PV output curves were then calibrated based on the average utilization hours of the 50 representative points.

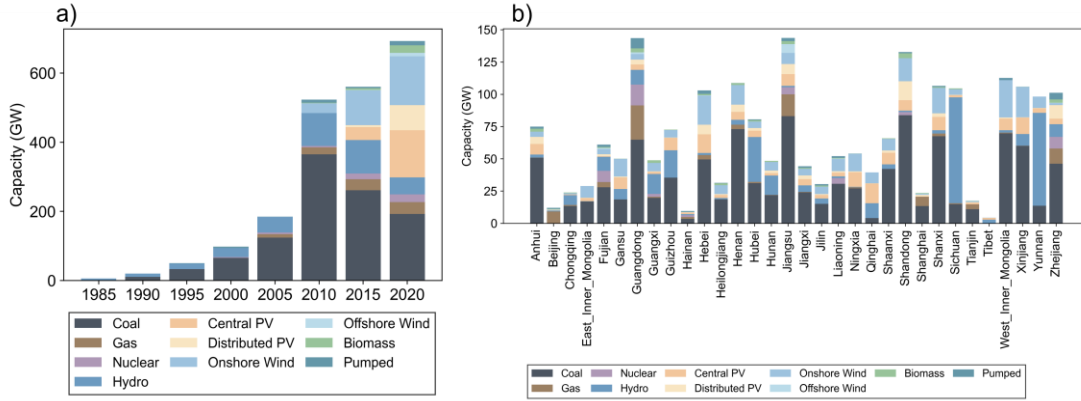
Since the wind and solar output curves differ across various regions, a more uniform spatial distribution helps to smooth the fluctuations in wind and solar power generation, though it may slightly reduce the average capacity factor. To model this, we constructed a function between the area selection weight and the capacity factor for each point, expressed as $\omega = CF^{-\alpha}$, where ω is the area selection weight, CF is the annual capacity factor, and α is the spatial distribution factor. When α is 0, it represents uniform distribution of wind and solar across all available areas. As α increases, the distribution becomes more uneven, favoring areas with higher CF, which results in higher average CF, increased volatility, and reduced available capacity potential for each province. In this study, we set $\alpha = 4$. By weighting each grid according to ω , we can obtain the available potential and representative output curves for each technology in each province. Finally, we selected 12 typical days to capture the variations in seasonal and weather conditions which significantly impact wind power generation. These 12 typical days represent various combinations of solar and wind output levels, reflecting the uncertainties inherent in solar and wind power generation.

(3) Historical power capacity

The historical installed capacity of various energy technologies has been predetermined in the model. This implies that transitioning from a system dominated by coal-fired power generation to one with a greater share of renewable energy requires consideration of additional costs for new investments, as well as the risk for asset stranding. The model considers the years up to 2020 as historical years, during which the installed capacity of different energy technologies has been documented.

For coal-fired power generation, we rely on a comprehensive database of coal-fired power units in China, commonly used for policy evaluations¹⁷. This database includes geographical coordinates, capacity, and coal consumption data for all coal-fired power units from 1989 to 2018. However, it does not cover the installed capacity of coal-fired power plants in 2019 and 2020. To address this, we adjust the installed capacity from 2016 to 2018 using data from the China Electric Power Yearbook¹⁸, aligning it with the actual installed capacity from 2016 to 2020. This adjustment covers a total of 706 coal-fired power projects, providing the historical coal power capacity for each 5-year period. For natural gas and biomass power generation, we utilize the Global Oil and Gas Plant Tracker¹⁹ and Global Bioenergy Power Tracker²⁰ databases from Global Energy Monitor to summarize historical gas and biomass power generation in China over various periods. However, there is a notable discrepancy between the biomass power data in these databases and the data reported in China's statistical yearbooks. To resolve this, we calibrate the historical biomass power generation capacity for each period using installed capacity data from provincial statistical yearbooks.

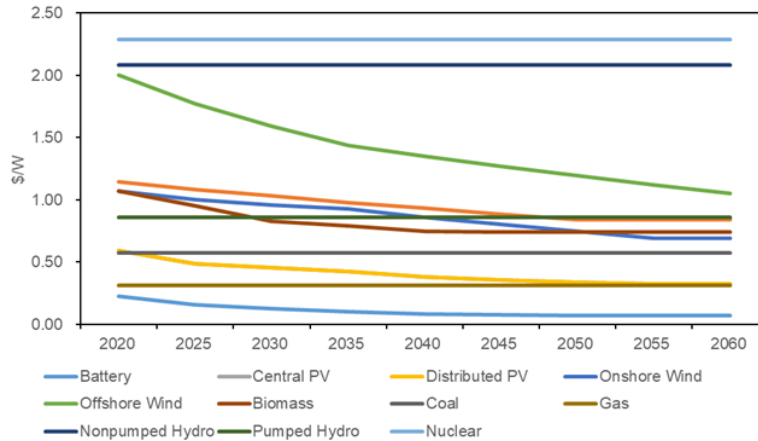
For nuclear power generation, we use data from the China Electric Power Yearbook and the Nuclear Power Basic Database (see https://www.smnpo.cn/dbdata/china_query.do) to summarize the new installed capacity of nuclear power in eight coastal provinces for various periods. Pumped storage and non-pumped hydroelectric units are similarly summarized using data from the Electric Power Yearbook for various periods. In addition, we exogenously set the pumped storage capacity plan in the next fifteen years according to the data collected by International Energy (see <https://www.in-en.com/article/html/energy-2330090.shtml>) to consider the growth of pumped storage capacity. As for offshore wind power, the cumulative capacity as of the end of 2015 was 1015MW. Due to a lack of information, we allocate the installed capacity from 2011 to 2015 to Fujian, Jiangsu, and Guangdong provinces based on the proportion of new installed capacity in 2015. The installed capacity from 2016 to 2020 is determined based on the cumulative installed capacity disclosed by the China Wind Energy Association (CWEA) for each province by the end of the 13th Five-Year Plan period. Onshore wind power is summarized using data from the Electric Power Yearbook. For central and distributed PV power generation, we summarize the annual provincial centralized and distributed photovoltaic power generation capacity disclosed by the National Energy Administration for the periods from 2011 to 2020. The historical power capacity mix is depicted in Supplementary Fig. 4.



Supplementary Fig. 4 Historical capacity installation by period and region. **a** national capacity installation by year, **b** provincial-level capacity mix in 2020. PV solar photovoltaic technology, GW gigawatt.

(4) Power generation technology costs

The costs of renewable energy generation technologies have been rapidly decreasing in recent years, and this trend is expected to continue in the future. We set the investment costs for various technologies using multiple open-source data sources. The investment cost data for battery storage technology is from Peng et al.⁴. The investment cost data for solar PV in 2025 and 2030 are derived from the 2022-2023 China PV Industry Development Roadmap compiled by the China PV Industry Association²¹, while the cost decline rate after 2030 is obtained from the median scenario of the INNOPATHS database²². For historical investment cost data for wind and solar power, we estimate the historical cost reduction rates for each period in China based on the International Renewable Energy Agency's (IRENA) analysis²³. The investment costs for coal-fired, nuclear, gas-fired, biomass, and hydropower generation are set based on recent information available in China, with the investment cost reduction rate for biomass power generation determined based on the median scenario of the INNOPATHS database. The investment costs for each technology are depicted in Supplementary Fig. 5. The fixed O&M cost is 20.4, 6.4, 42.2, 10.6, 100.0, 20.4, 13.1, 21.3, 5.6, 6.6 USD·MW⁻¹·year⁻¹ for coal, gas, non-pumped hydro, pumped hydro, nuclear, biomass, onshore wind, offshore wind, central PV and distributed PV in the base year, respectively^{3,6}. The cost reduction rate of fixed O&M is the same as the investment cost. The startup cost of coal-fired, gas-fired and biomass-fired power is set as \$150, \$42 and \$70/MW⁶.



Supplementary Fig. 5 Investment costs of power generation and energy storage technologies.

The unit of battery cost is dollars per watt-hour (\$/Wh). PV solar photovoltaic technology, \$/W dollars per watt.

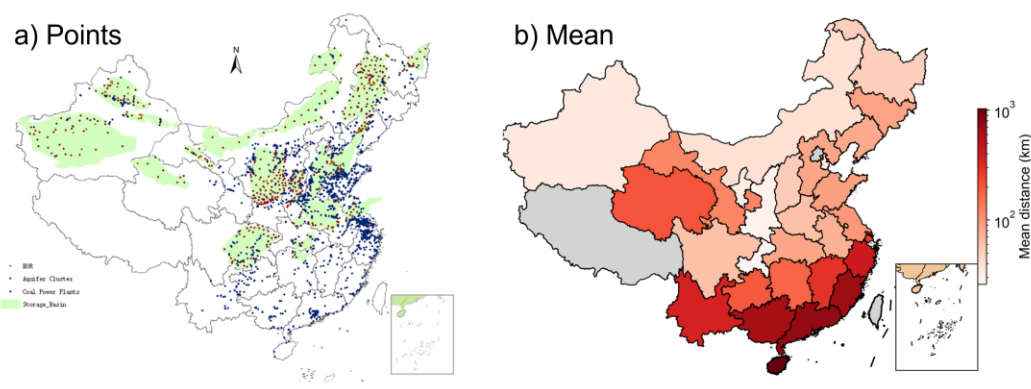
(5) CCS technology cost

Carbon capture and storage (CCS) retrofitting involves several types of costs. The first is related to the investment cost. Koelbl et al. (2014)²⁴ estimated the magnitude and uncertainty of additional investment costs for CCS in coal-fired, natural gas-fired, and biomass-fired power generation. Their estimates indicate that the investment costs for coal-fired power plants without retrofitting in 2020 ranged from \$749 to \$2939 per kilowatt, while the additional CCS investment costs ranged from \$219 to \$1212 per kilowatt. For biomass-fired power plants, the investment costs without retrofitting ranged from \$1161 to \$3251 per kilowatt, with additional CCS investment costs ranging from \$548 to \$902 per kilowatt. For natural gas-fired power plants, the investment costs without CCS ranged from \$436 to \$949 per kilowatt, while the additional CCS costs ranged from \$266 to \$1013 per kilowatt. Additionally, Rubin et al.²⁵ estimated the investment costs for different capture technologies, suggesting a 74% increase in investment costs for coal-fired power generation and a 96% increase for natural gas-fired power generation.

Due to significant differences in technology investment costs in China compared to the studies mentioned above, we recalculated the unit investment costs for CCS based on the actual investment costs for coal-fired, natural gas-fired, and biomass-fired power generation, as well as the proportion of CCS-related investment costs. Additionally, CCS technology incurs both fixed and variable operation and maintenance (O&M) costs. According to the findings of Morris et al.²⁶, we set the fixed O&M costs for CCS in coal-fired power plants in 2020 at \$0.023 per watt per year and the variable O&M costs at \$2.2 per megawatt-hour. For natural gas-fired power plants, these costs were set at \$0.029 per watt per year and \$3.7 per megawatt-hour, while for biomass-fired power plants, they were set at \$0.060 per watt per year and \$3.3 per megawatt-hour. In addition to O&M costs, CCS also leads to a loss of power output from power plants. We set the energy penalties for coal-fired, gas-fired, and biomass-fired power generation at 24%^{27,28}, 18%^{28,29}, and 51%³⁰, respectively, in 2025, with a gradual reduction in these penalties due to technological advancements³¹. The carbon capture rate for coal-fired, gas-fired, and biomass-fired power plants is set at 90%³².

The carbon dioxide captured must be transported to storage sites. Based on existing research, we considered two types of storage sites: saline aquifers and enhanced oil recovery (EOR) in oil fields, with relevant storage area data sources cited^{33,34}. For saline aquifers, we generated county-level storage points, excluding certain counties based on prohibition indicators such as geological faults, seismic activity, population density, surface water, national parks, national forest parks, and national nature reserves³⁵. The locations of coal-fired power plants and the two types of storage points are shown in Supplementary Fig. 6. Since the model operates at the provincial level, we averaged the carbon transport distances to the provincial level. Using the spatial distribution of coal-fired power plants, we calculated the distances to the nearest CCS storage points and performed a weighted average at the regional level based on plant capacity to obtain the average CCS transport distance for each province. The transportation costs vary depending on the method used. We considered pipeline transport, shipping, and railway tank wagon transport, calculating the average transportation cost per ton of carbon per kilometer, set at 0.026 USD·t⁻¹·km⁻¹³⁶. Combining the average transportation distances for each province, carbon transportation costs range from \$0.7 to \$27 per ton of CO₂ across provinces. Provinces in the western and northern regions, such as Chongqing and Shaanxi, have the lowest transportation costs, while provinces in the southern

regions, such as Hainan, Guangdong, and Fujian, have the highest. Clearly, CCS transportation costs will significantly limit the feasibility of implementing CCS in coal-fired power plants across different provinces. The carbon storage cost was set at \$5.0 per ton of CO₂, as referenced from the annual report on carbon capture, utilization, and storage (CCUS) in China³⁷. See Supplementary Table 7 for further details on CCS technology costs and operation parameters. Due to the high uncertainty surrounding CCS technology in the future, this study sets the costs and operational parameters for CCS under a low technological advancement scenario, as shown in Supplementary Table 8, which serves as a key scenario for sensitivity analysis.



Supplementary Fig. 6 Provincial-level carbon storage availability. **a** points of existing coal-fired power units and storage; **b** averaged transport distance of each province. EOR enhanced oil recovery, km kilometer.

Supplementary Table 7 Carbon capture retrofitting costs and operation parameters of thermal power

Parameters	Tech	2025	2030	2040	2050	2060
Carbon capture retrofit investment cost (\$/kW)	Coal	381.9	305.5	195.5	158.4	142.5
	Gas	271.5	217.2	139.0	112.6	101.4
	Biomass	716.0	572.8	366.6	296.9	267.2
Fixed O&M cost (\$·kW ⁻¹ ·year ⁻¹)	Coal	20.7	16.6	10.6	8.6	7.7
	Gas	26.1	20.9	13.4	10.8	9.7
	Biomass	54.0	43.2	27.6	22.4	20.2
Carbon transport cost (\$·t ⁻¹ ·km ⁻¹)	/			0.026 (0.020-0.036)		
Carbon storage cost (\$/t)	/			5.0 (3.0-8.5)		
Carbon capture rate (%)	/			90		
Energy penalty (%)	Coal	24.4	22.2	15.6	13.3	11.1
	Gas	18.3	16.7	11.7	10.0	8.3
	Biomass	51.3	46.7	28.0	24.0	20.0

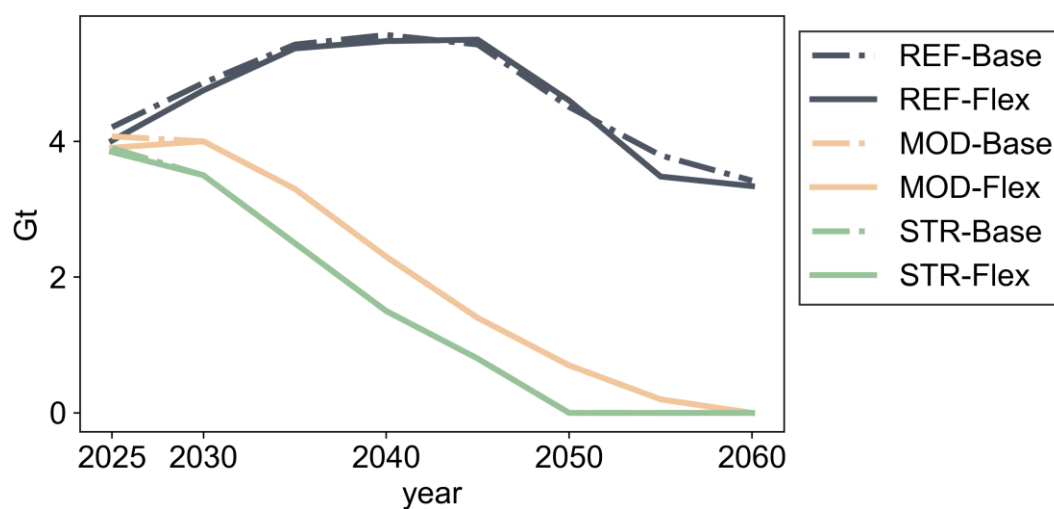
Note: \$ dollars, kW kilowatt, t ton, km kilometer.

Supplementary Table 8 Carbon capture retrofitting costs and operation parameters of thermal power under low technological advancement assumption

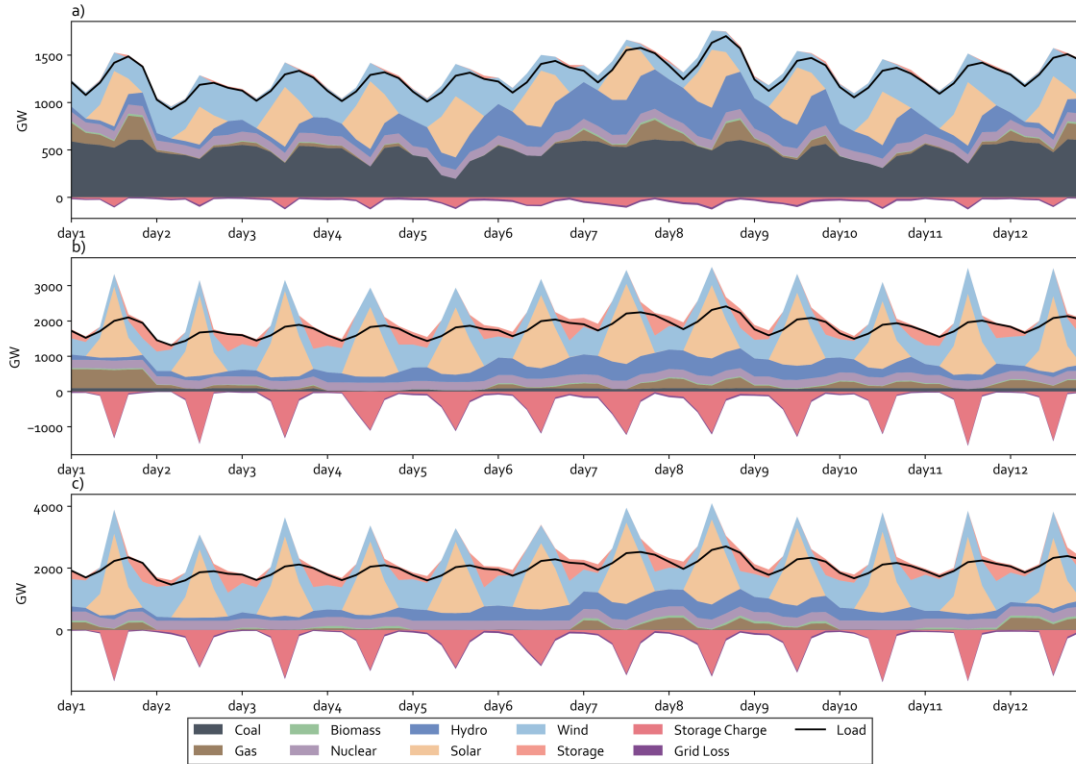
Parameters	Tech	2025	2030	2040	2050	2060
Carbon capture retrofit investment cost (\$/kW)	Coal	424.3	381.9	309.3	309.3	309.3
	Gas	23.0	271.5	219.9	219.9	219.9
	Biomass	795.5	716.0	580.0	580.0	580.0
Fixed O&M cost (\$·kW ⁻¹ ·year ⁻¹)	Coal	23.0	20.7	16.8	16.8	16.8
	Gas	301.7	26.1	21.1	21.1	21.1
	Biomass	60.0	54.0	43.7	43.7	43.7
Carbon transport cost (\$·t ⁻¹ ·km ⁻¹)	/			0.036		
Carbon storage cost (\$/t)	/			10		
Carbon capture rate (%)	/			90		
Energy penalty (%)	Coal	24.4	24.1	23.8	23.5	23.2
	Gas	18.3	18.1	17.9	17.6	17.4
	Biomass	51.3	49.7	48.1	46.5	44.9

Note: \$ dollars, kW kilowatt, t ton, km kilometer.

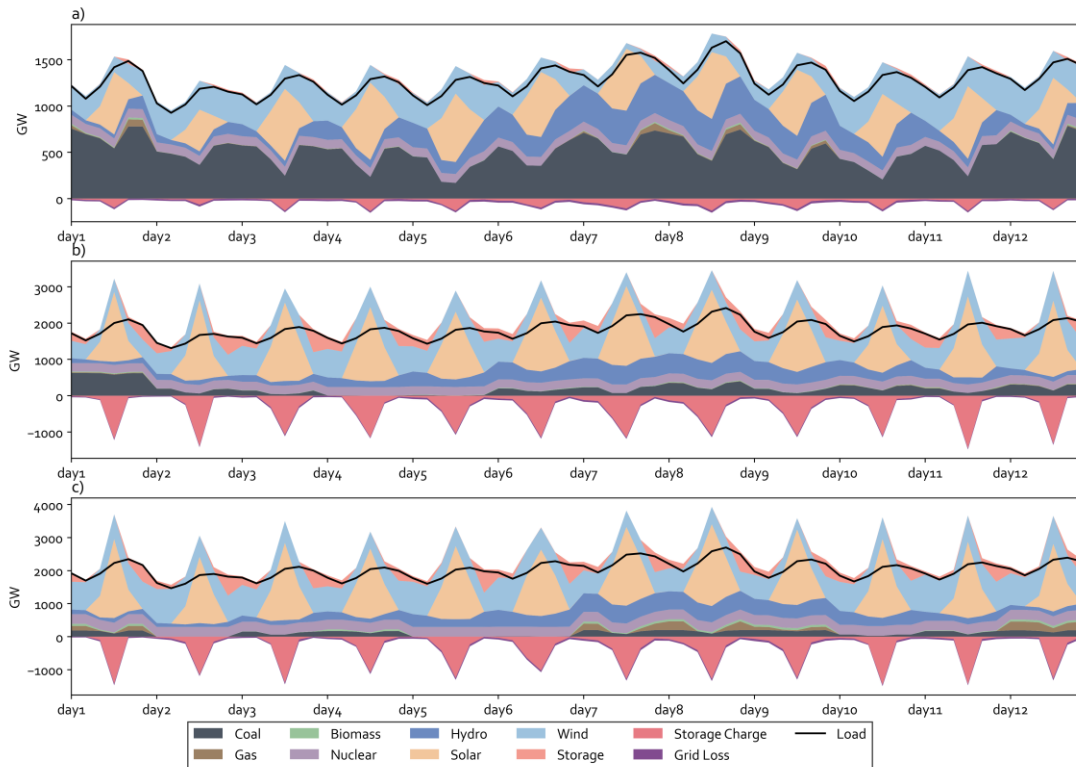
Supplementary Figures



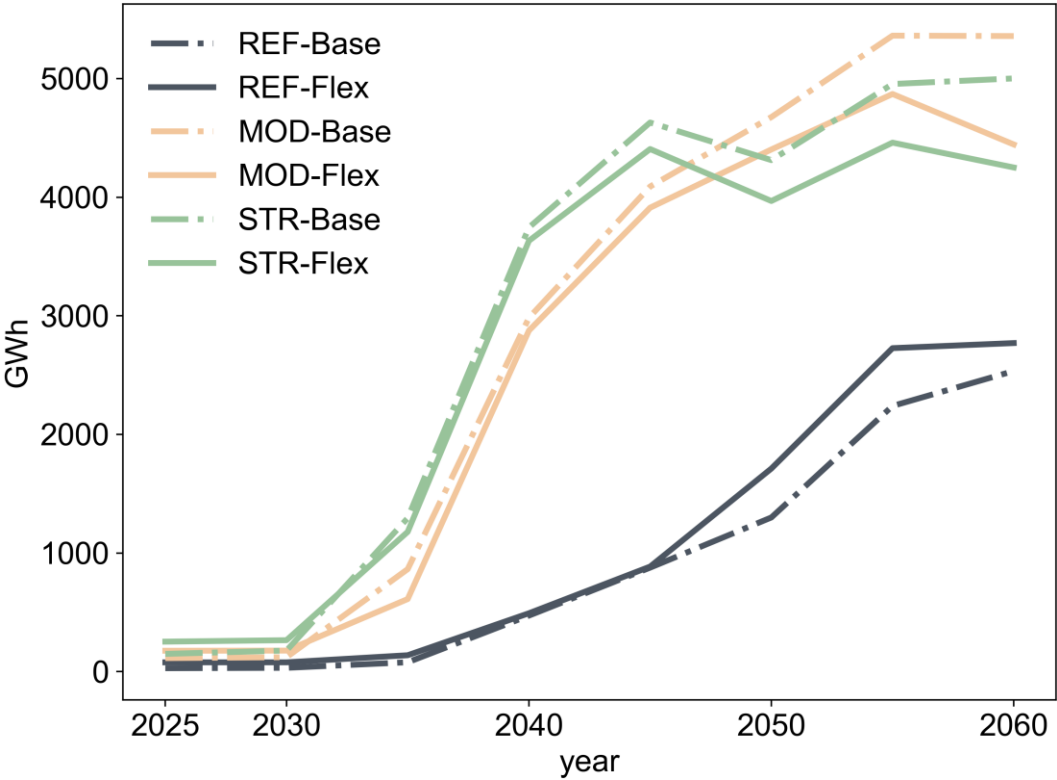
Supplementary Fig. 7 Emission trajectories of six scenarios. REF, MOD and STR represent the reference, moderate and strict carbon emission caps on power sector; Flex and Base represent that coal power can be dispatched flexibly and operates as a base load, respectively. Gt gigaton.



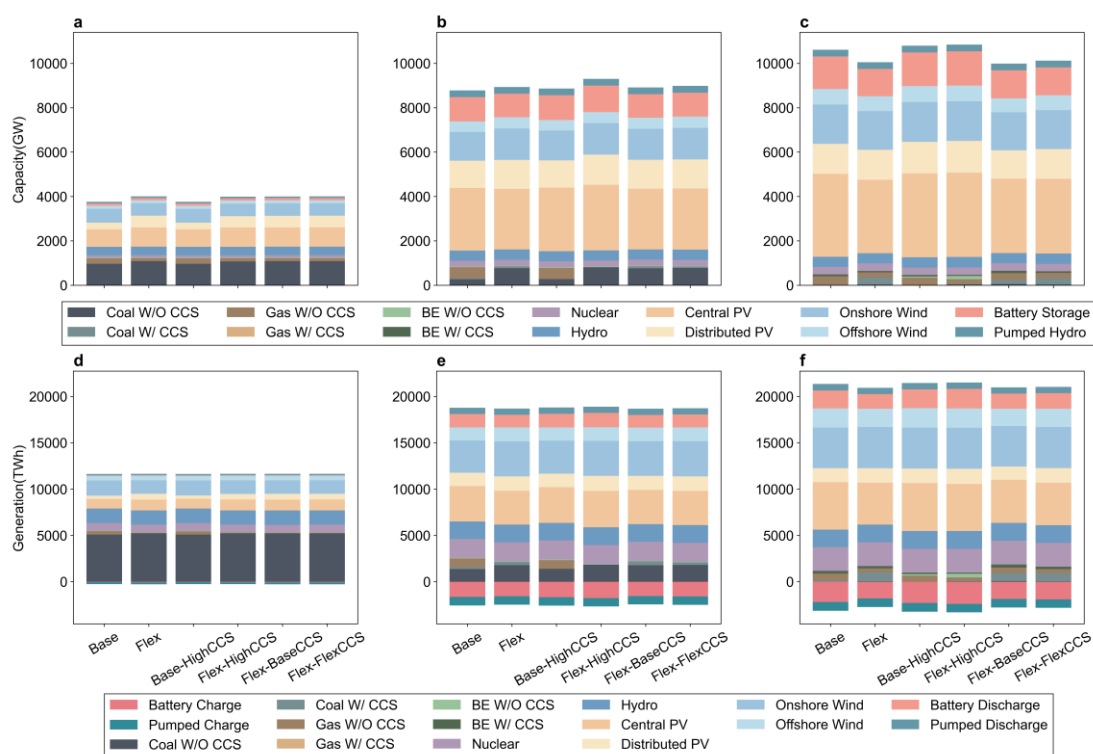
Supplementary Fig. 8 National power system dispatch curves in three years under STR-Base scenario. a 2020, b 2030, c 2060. STR represents stringent emission cap, Base represents that coal power operates as a base load. GW gigawatt.



Supplementary Fig. 9 National power system dispatch curve in three years under STR-Flex scenario. a 2020, b 2030, c 2060. STR represents stringent emission cap, Flex represents that coal power can be dispatched flexibly. GW gigawatt.



Supplementary Fig. 10 Battery energy storage capacity in six scenarios. REF, MOD, and STR represent no, moderate and stringent emission caps, respectively; Flex and Base represent that coal power can be dispatched flexibly and operates as a base load, respectively. GWh gigawatt-hour.



Supplementary Fig. 11 National power system capacity and generation mix under different sensitivity experiments. a capacity mix in 2030; b capacity mix in 2045; c capacity mix in 2060; d generation mix in 2030; e generation mix in 2045; f generation mix in 2060. Scenario names at the X-axis are combinations of two dimensions. The emissions trajectory of each scenario is set as the moderate emissions cap. Flex and Base represent that coal power can be dispatched flexibly and operates as a base load, respectively. HighCCS indicates higher CCS cost, BaseCCS indicates that CCS-retrofitted coal power can only operate as a base load, FlexCCS indicates that CCS-retrofitted coal power can be further retrofitted to be dispatched flexibly with additional investment costs. PV solar photovoltaic technology, CCS carbon capture and storage, W/ with, W/O without, GW gigawatt, TWh terawatt-hour. Detailed descriptions can be found in Supplementary Table 9.

Supplementary Tables

Supplementary Table 9 Description of sensitivity analysis scenarios

Flex scenario	Base scenario	Description
LowFlex	Base	Lower flexibility compared to MOD-Flex; Minimum output level is 0.5
Flex	Base	MOD-Flex
HighFlex	Base	Higher flexibility compared to STR-Flex; Minimum output level is 0.3
Flex-MidREPot	Base-MidREPot	Mid VRE potential as shown in Supplementary Table5 and Supplementary Table 6
Flex-HighREPot	Base-HighREPot	High VRE potential as shown in Supplementary Table5 and Supplementary Table 6
Flex-	Base-	Lower VRE investment cost. PV cost is \$0.40, \$0.31 and

LowRECost	LowRECost	\$0.25/W; onshore wind cost is \$0.63, \$0.47 and \$0.38/W; offshore wind cost is \$1.36, \$1.10 and \$1.05/W in 2030, 2045 and 2060, respectively.
Flex-LowBatteryCost	Base-LowBatteryCost	Battery capacity cost is lower, i.e., \$1.03, \$0.63 and \$0.57/Wh.
Flex-LowREBatCost	Base-LowREBatCost	Both lower VRE and battery costs. Other parameters are the same as the Flex scenario
Flex-HighCoal	Base-HighCoal	Higher coal price, a 5% growth rate per 5 years since 2020.
Flex-LowGas	Base-LowGas	Lower gas price, a 5% decline rate per 5 years since 2020
Flex-LowCoal	Base-LowCoal	Lower coal price, kept in the level of 2019
Flex-HighDemand	Base-HighDemand	Higher electricity demand. A 0.25% growth rate per year since 2020
Flex-LowDemand	Base-LowDemand	Lower electricity demand. A 0.25% decline rate per year since 2020
Flex-HighCCS	Base-HighCCS	Higher CCS cost. Costs and parameters are set at Supplementary Table 8
Flex-BaseCCS	Base	CCS-retrofitted coal power serves as a base load. Annual utilization of CCS-retrofitted coal power must exceed 5000h for any period
Flex-FlexCCS	Base	CCS-retrofitted coal power can be retrofitted to be dispatched flexibly. The investment cost of CCS increases by 20% for any period.

Note: the emissions trajectory of each scenario is the moderate emission cap scenario. To ensure consistency in the sensitivity analysis, Flex scenarios correspond to Base scenarios in the same row, allowing for a direct comparison between the two scenarios. Flex and Base represent that coal power can be dispatched flexibly and operates as a base load, respectively.

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